VIDYALANKAR INSTITUTE OF TECHNOLOGY

ReviewIQ

AI-Based Product Review Intelligence and Consumer Insight System

NLP · Deep Learning · Advanced ML

PROJECT TEAM

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Guide: Prof. Sneha Annappanavar

Problem and Objectives

A popular product can have thousands of free-text reviews that nobody can read in full. Buyers ask **"should I buy this?"**; sellers ask **"what should we improve?"**

- Classify review sentiment (positive, neutral, negative) with NLP, deep learning and advanced ML models
- Rank complaint areas by aspect (battery, sound, camera)
- Score review helpfulness and flag suspicious reviews
- Answer product questions from reviews, with evidence
- Deliver everything in one interactive dashboard

Dataset and Preprocessing

50,000 Amazon Electronics reviews

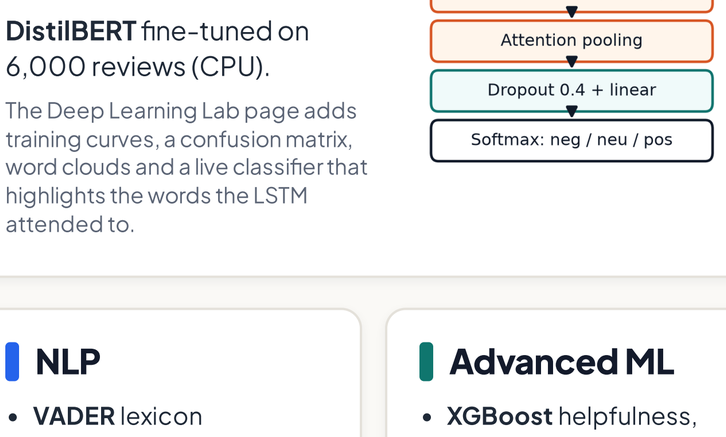
34,365 products

8,883 reviewers

80 / 7 / 13 % positive / neutral / negative

Amazon Reviews 2023, cleaned and labelled from star ratings (4–5 positive, 3 neutral, 1–2 negative). Product name, brand and image joined by parent ASIN. One shared 40,000 / 10,000 train and test split.

System Architecture



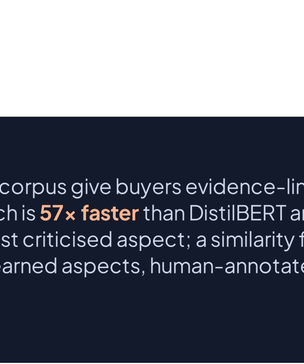
Reviews and metadata feed three branches, then product intelligence and grounded question answering.

Deep Learning Branch

BiLSTM + Attention trained from scratch: 14,937–word vocabulary, 2 x 128 bidirectional LSTM, 2.57 M parameters, class-weighted loss, about 7 minutes on a laptop GPU.

DistilBERT fine-tuned on 6,000 reviews (CPU).

The Deep Learning Lab page adds training curves, a confusion matrix, word clouds and a live classifier that highlights the words the LSTM attended to.



NLP

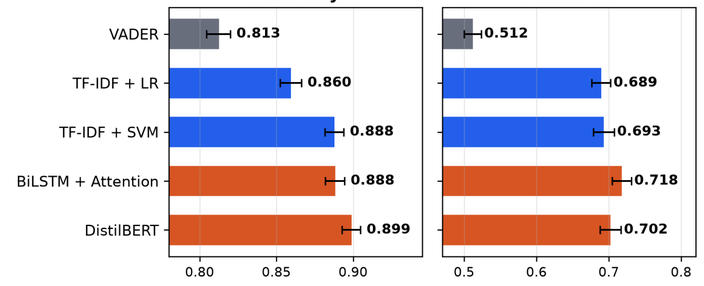
- **VADER** lexicon
- **TF-IDF** + LR / SVM
- **LDA**, 8 topics
- **Aspect sentiment** with evidence
- **MiniLM** search embeddings

Advanced ML

- **XGBoost** helpfulness, 70.5% accuracy
- **MiniBatchKMeans**, k = 6
- **Isolation Forest**, 10% flagged

Sentiment Results (10,000 test reviews)

Model	Accuracy	Macro-F1	F1 neutral
VADER	.813	.512	.080
TF-IDF + LR	.860	.689	.400
TF-IDF + SVM	.888	.693	.382
BiLSTM + Attention	**.888**	**.718**	**.446**
DistilBERT	**.899**	**.702**	**.368**



95% bootstrap intervals. BiLSTM beats the SVM on macro-F1 (+0.025, significant); DistilBERT is best on accuracy.

Ablation: What Drives the Gain?

LSTM variant (3 seeds, mean)	Accuracy	Macro-F1
Full (attention + class weights)	**.891**	**.722**
No class weights	**.899**	.709
Mean pooling instead of attention	.889	.715

Class weighting, **not attention**, drives most of the macro-F1 gain. Without it the LSTM matches DistilBERT's accuracy (**.899**).

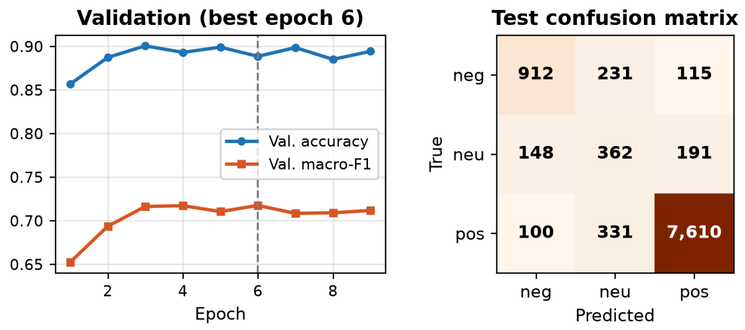
Cost and Speed

26× fewer parameters than DistilBERT

57× faster on CPU (975 vs 17 reviews/s)

13× faster on GPU

Training and Errors



155 of the LSTM's 1,116 test errors are confident, and many are **label noise**: the stars disagree with the text.

5 stars, predicted negative: "Piece of junk !!! Poor quality, high price .. it lasted 10weeks"

1 star, predicted positive: "Worked out of the box. love it"

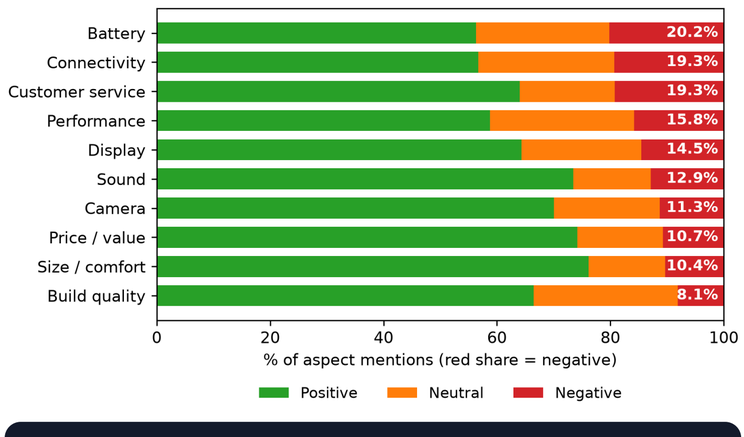
What Each Sentiment Sounds Like



Negative Positive

Words weighted by how characteristic they are of each class: "return", "waste", "money" versus "easy", "love", "perfect".

Aspect Intelligence



Battery is the top complaint (20.2% negative mentions), then **connectivity** and **customer service** (19.3% each).

Grounded Question Answering

Product-scoped retrieval (MiniLM, top 5 reviews) plus FLAN-T5, with a **similarity floor**: weak evidence means "not enough information".

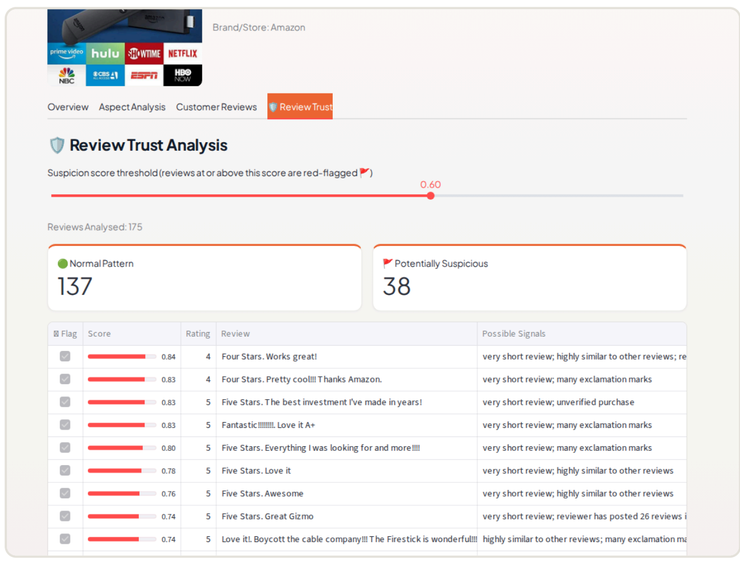
97.8% answerable questions accepted

95.6% unanswerable questions declined

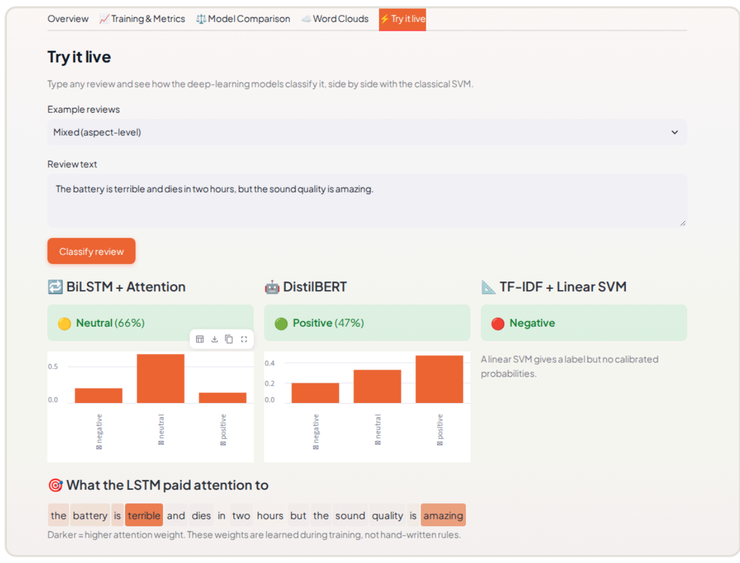
100% off-topic questions declined

Semantic retrieval beat TF-IDF on paraphrased aspect queries (0.54 vs 0.32 aspect-mention rate at 5).

Interactive Dashboard



Review Trust: unsupervised Isolation Forest flags 4,999 reviews (10%) as potentially suspicious, with a threshold slider and reasons per review. No fake-review accuracy is claimed.



Deep Learning Lab: live classifier with attention

Limitations

- Labels come from star ratings, so some are noisy
- Sample is a reviewer-grouped block of one category
- DistilBERT saw one-sixth of the LSTM's data
- Flagged reviews are anomalies, not confirmed fakes

CONCLUSION

NLP, deep learning and advanced ML on one corpus give buyers evidence-linked insight and sellers a ranked list of problems. A BiLSTM trained from scratch is **57× faster** than DistilBERT and matches its accuracy once class weights are removed; **battery** is the most criticised aspect; a similarity floor stops unsupported answers. **Future work:** full-data DistilBERT, learned aspects, human-annotated subsets, random multi-category sampling.

KEY REFERENCES

[1] Hutto and Gilbert, VADER, ICWSM 2014. [2] Hochreiter and Schmidhuber, LSTM, 1997. [3] Sanh et al., DistilBERT, 2019. [4] Chen and Guestrin, XGBoost, KDD 2016. [5] Liu et al., Isolation Forest, ICDM 2008. [6] Reimers and Gurevych, Sentence-BERT, EMNLP 2019. [7] Lewis et al., RAG, NeurIPS 2020. [8] Hou et al., Amazon Reviews 2023, 2024.

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